

SPE Selection Technique: A Projection-based Approach for Precise Object Interaction in Immersive Virtual Environments

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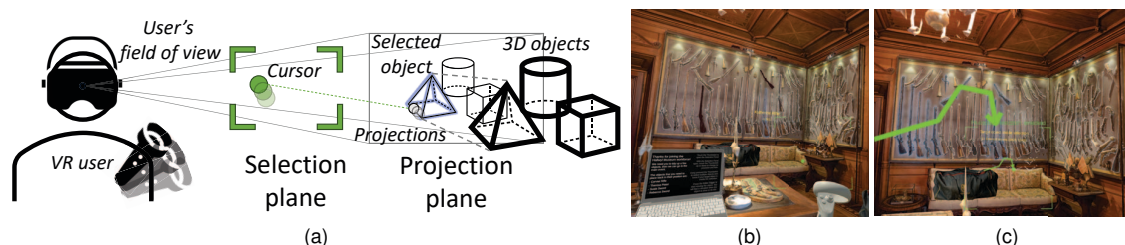


Figure 1: The SPE technique: a) interaction paradigm props, b) example of a single-object selection, and c) of a multiple selection.

ABSTRACT

Selecting objects has always represented one of the fundamental tasks required by the majority of immersive applications. Despite the key role played by selection techniques in the design of immersive experiences, there is not a dominant or a one-size-fits-all approach yet. In this scenario, the current paper proposes a novel selection technique, named Selection among Projected Elements, which focuses on the selection of distant, small and possibly close, objects. The devised approach was assessed in the context of a virtual heritage application that was specifically designed with the aim of stressing the capabilities of the SPE technique.

1 INTRODUCTION

According to [7], a well-designed selection technique should be (i) intuitive, (ii) support a seamless usage, and (iii) be robust to possible errors made by the users in order to boost productivity and user acceptance, as well as to reduce potential fatigue. In addition to the above concerns, the design of selection techniques for immersive experiences needs to take into account difficulties related to the higher degree of freedom offered by 3D environments compared to 2D ones, which can lead to the lack of precision and the increase of users' frustration [7].

Given the various requirements that can be encountered in the development of VR applications (e.g., the need to use the hands/the controllers' buttons for tasks different than selection, to operate onto multiple objects at the same time, to move while grabbing them, etc.), it is not surprising that a single selection technique did not emerge as the ultimate solution for all the possible scenarios.

In this paper, the focus was posed on aspects regarding the selection of (and the interaction with) multiple objects that can be distant from the users or simply small-sized. In particular, the paper proposes a novel selection technique named Selection among Projected Elements, abbreviated SPE, which builds upon a previous

interaction method, i.e., the Precise and Rapid Interaction through Scaled Manipulation, or PRISM [1]. The main idea of the proposed technique is to leverage the projections that selectable objects make on the user's image plane. By simulating a cursor-based interaction of a 2D user interface (UI), the user can move a cursor with his or her hand/the controller and select one or more objects. To deal with inaccuracies possibly experienced during the selection of distant/small and close objects, the sensitivity of cursor movements can be manually adjusted by the user, facilitating precise interactions.

In order to assess the effectiveness of the devised approach, a VR application was designed in which the users have to interact with a number of artworks exposed in a museum through the SPE technique and solve an easy find-and-place game.

2 RELATED WORK

The ray cast paradigm is indeed one of the most popular selection methods, thanks to its simplicity and generality [2]. However, it shows a number of limitations when it is used to select small-sized and/or objects that are far from the user [7]. In particular, the effectiveness of this method decreases with the distance from the targets, as a small change in the orientation of the pointing element translates into large movements of the casted ray [3]. Works in literature reported a number of techniques particularly suitable for selecting distant/small and close objects like, e.g., the GO-GO [5], Word-In-Miniature (WIM) [6], PRECIOUS [3] and Sphere-casting refined by QUAD-menu (SQUAD) [2].

The proposed SPE technique looks at the benefits possibly coming from the use of the PRISM metaphor [1] to overcome the above limitations. Originally designed for manipulating objects in 3D environments, the PRISM technique lets the user dynamically change the ratio between the movements of the hands/controllers and the transformation properties of the object to be controlled (i.e., its translations and/or rotations). In particular, the object transformations can be manipulated with a precision that changes depending on the hand's/controller's movement speed. As the speed of the input element decreases, the precision of the manipulation increases, thus obtaining fine-grained transformations. To reduce the possible instability related to the tracking of the input element and increase the precision, the applied transformations are projected to an imaginary plane that lets the user adjust only two degrees of freedom at a time. This aspect of the PRISM technique recalls cursor-based interactions used in 2D UIs. The SPE technique transforms the above metaphor

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to let the user select multiple objects projected onto an imaginary plane with a virtual cursor that can be manipulated with controllable scaling factors.

The use of a projection plane in the SPE technique presents similarities with the four image plane-based interaction methods (namely, Head Crusher, Sticky Finger, Lifting Palm, and Framing Hands) proposed in [4], as the user interacts with 3D elements by leveraging their 2D projection on the user's image plane. The four methods differ from each other in the articulation of the hand(s) during the selection process. Differently than in the above methods, though, the SPE technique provides a mechanism to alter the mapping between the input element (the controller) and the position of the virtual cursor leveraged for interacting with the objects.

3 THE SPE TECHNIQUE

The proposed technique lets the user select elements from a dense and distant set of objects. The interaction paradigm (illustrated in Fig. 1a) relies on the steps described below.

The user activates the selection by touching the controller's Thumbstick. All the objects included in the user's field of view are automatically added to the list of selectable objects and highlighted with a white outline (as shown in Fig. 1b). The positions of the selectable objects are projected (but not visualized) onto an invisible plane (named *projection plane*) that is located between the user and the objects at a predefined distance along the user's view direction. To enable the cursor-based selection paradigm and make it possible to adjust the scaling factor, the positions of the objects in the projection plane are remapped onto a new plane, named *selection plane* and showed as dots on it. The plane is shown to the user with a green frame (Fig. 1b). The size of the selection plane, which can be adjusted by moving the Thumbstick up or down to respectively increase or decrease it, determines the precision of the interaction: the larger the selection plane, the greater is the control of the user over the cursor (the green dot in Fig. 1b), resulting into finer operations. The position of the cursor is controlled by the movements of the controller. When the user moves the controller, the system computes the distance between the cursor and the position of all the selectable objects remapped onto the selection plane. The closest object is highlighted and automatically added to the selection set. It is possible to perform a multiple selection and add a given object to the set by pressing (not just touching it, and keeping it pressed) the Thumbstick and moving the cursor within the same selection session; a session ends when the user fully releases the Thumbstick. An example of a multiple selection is depicted in Fig. 1c.

Once selected, the objects can be moved by using the original PRISM technique [1] adapted for immersive virtual environments. When transformations are applied to the objects, the system needs to recalculate their projections and remapped positions onto the projection and selection plane, respectively.

4 USE CASE

In order to show the benefits of the proposed interaction technique, a virtual heritage VR application was considered as a use case. In particular, the users are requested to interact with the artworks belonging to Wilhelmina von Hallwyl's collection of antique arms and armors (source of the 3D model <https://bit.ly/3JJChK>) playing a simple serious game. The goal of the game is to find given target objects in the environment and grab them to a specific location. Once the user has selected one or more objects, information regarding the artworks is shown, reporting the most relevant historical data. To increase the difficulty of the game, the time available for completing the task is decreased at each trial. A video showing a user interacting with the application and the executable itself are available for download respectively at https://tiny.cc/spe_3dui_vid and https://tiny.cc/spe_3dui_build. This use case was chosen since the set of objects the user is requested to interact with

includes more than 160 elements distributed densely in the showcase room. Moreover, in this use case, the multi-selection feature of the SPE technique can be exploited to speed up the search of relevant objects. To this aim, the game shows a message informing the user that the current selection contains the target object to be moved for completing the current task. In this way, the user can iteratively select a single showcase from those in the room and focus the search on it (if the target object is within the current selection), without the need to read all the descriptions of objects in the others showcases.

5 CONCLUSION AND FUTURE WORK

This paper describes the SPE technique, which is meant to support the selection of distant/small-sized objects. The proposed technique was experimented in the context of virtual heritage application for easing the interaction with artworks while playing a simple serious game. From the preliminary open-feedback collected, it emerged that the SPE technique provided the users with good control over the selection of dense objects. The dot-based representation on the selection plane was found as particularly helpful; for instance, when the user was attempting to select a given object in a dense region, he or she immediately received a feedback on the corrective movement required to reach the intended object. From the perspective of ergonomics, the positioning of the selection plane was found as not straining and enabling, thanks to its adjustable size, a wide-spanning over the set of selectable objects with small hand movements. The fact of having all the key operations mapped on the same button was judged as intuitive and optimized, since many unused buttons are still available on the controllers if required by the application scenario. Some users stated that it could have been nice to complement the multi-selection with a box-like group selection. Besides considering suggestions collected so far, future work could be aimed at evaluating the proposed technique from different perspectives, e.g., performance, usability, and effectiveness, through user studies involving other scenarios and tasks, and the collection of structured subjective feedback as well as objective data. Moreover, two-hand based interactions could be investigated, e.g., for dynamically defining the size of the projection plane or enabling a new way to manage multi-selection.

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